

Case Reports > [Neurol India](#). 2023 Jan-Feb;71(1):86-91. doi: 10.4103/0028-3886.370449.

Post-Vaccination/Post-COVID Immune-Mediated Demyelination of the Brain and Spinal Cord: A Novel Neuroimaging Finding

Ashima Mahajan¹, Manoj K Nayak¹, Shailesh B Gaikwad¹, Kalyan Sharma¹,
M Vasantha Padma Srivastava², Pooja Anand², Rahul Oinam², Biswa Mohan Mishra²

Affiliations + expand

PMID: 36861579 DOI: [10.4103/0028-3886.370449](#)

Free article

Abstract

Introduction: Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) causes coronavirus disease 2019 (COVID-19), which not only produces respiratory symptoms but is known to involve almost every system, and its neuroinvasive properties have been well demonstrated throughout the pandemic. Also, to combat the pandemic, there was rapid development and induction of various vaccination drives, following which many adverse events following immunization (AEFIs) have been reported, which include neurological complications as well.

Method: We present a series of three cases, post vaccination, with and without a history of COVID illness that showed remarkably similar findings on magnetic resonance imaging (MRI).

Result: A 38-year-old male presented with complaints of weakness of the bilateral lower limbs with sensory loss and bladder disturbance a day after receiving his first dose of ChadOx1 nCoV-19 (COVISHIELD) vaccine. A 50-year-old male with hypothyroidism characterized by autoimmune thyroiditis and impaired glucose tolerance experienced difficulty in walking 11.5 weeks after being administered with COVID vaccine (COVAXIN). A 38-year-old male presented with subacute onset progressive symmetric quadriparesis 2 months after their first dose of a COVID vaccine. The patient also had sensory ataxia, and his vibration sensation was impaired below C7. All three patients had typical pattern of involvement of the brain and spine on MRI with signal changes in bilateral corticospinal tracts, trigeminal tracts in the brain, and both lateral and posterior columns in the spine.

Conclusion: This pattern of brain and spine involvement on MRI is a novel finding and is likely a result of post-vaccination/post-COVID immune-mediated demyelination.

Keywords: COVID-19; COVID-19 vaccination; Leukoencephalopathy with brainstem and spinal cord involvement and lactate elevation (LBSL); demyelination; motor neuron disease (MND); neurotropism; sub-acute combined degeneration (SACD).

PubMed Disclaimer

Conflict of interest statement

None

Similar articles

[Seropositive Neuromyelitis Optica in a Case of Undiagnosed Ankylosing Spondylitis: A Neuro-Rheumatological Conundrum.](#)

Ghosh Md R, Roy D, León-Ruiz M, Das S, Dubey S, Benito-León J.
Qatar Med J. 2022 Jul 7;2022(3):29. doi: 10.5339/qmj.2022.29. eCollection 2022.
PMID: 35864917 [Free PMC article.](#)

[MOG encephalomyelitis after vaccination against severe acute respiratory syndrome coronavirus type 2 \(SARS-CoV-2\): case report and comprehensive review of the literature.](#)

Jarius S, Bieber N, Haas J, Wildemann B.
J Neurol. 2022 Oct;269(10):5198-5212. doi: 10.1007/s00415-022-11194-9. Epub 2022 Jun 23.
PMID: 35737110 [Free PMC article.](#) [Review.](#)

[Bilateral Immune-Mediated Keratolysis After Immunization With SARS-CoV-2 Recombinant Viral Vector Vaccine.](#)

Khan TA, Sidhu N, Khan L, Sen S, Hussain N, Tandon R, Gupta N.
Cornea. 2021 Dec 1;40(12):1629-1632. doi: 10.1097/ICO.0000000000002844.
PMID: 34483273 [Review.](#)

[Acute Transverse Myelitis \(ATM\):Clinical Review of 43 Patients With COVID-19-Associated ATM and 3 Post-Vaccination ATM Serious Adverse Events With the ChAdOx1 nCoV-19 Vaccine \(AZD1222\).](#)

Román GC, Gracia F, Torres A, Palacios A, Gracia K, Harris D.
Front Immunol. 2021 Apr 26;12:653786. doi: 10.3389/fimmu.2021.653786. eCollection 2021.
PMID: 33981305 [Free PMC article.](#)

[Leukoencephalopathy with brainstem and spinal cord involvement and lactate elevation \(LBSL\): assessment of the involved white matter tracts by MRI.](#)

Kassem H, Wafaie A, Abdelfattah S, Farid T.
Eur J Radiol. 2014 Jan;83(1):191-6. doi: 10.1016/j.ejrad.2013.09.023.
PMID: 24558666

[See all similar articles](#)

Cited by

[Cerebellar encephalitis and peripheral neuropathy with an atypical clinical and neuroimaging signature following Covid-19 vaccine: a report of two cases.](#)

Sicard M, Shor N, Davy V, Rouby JJ, Oquendo B, Maisonnobe T, Puybasset L, Lehericy S, Lecarpentier A, Donadio C, Oasi C, Belmin J, Lubetzki C, Corvol JC, Grabli D, Saracino D.
J Neurol. 2024 May 4. doi: 10.1007/s00415-024-12390-5. Online ahead of print.
PMID: 38704487 [No abstract available.](#)

[Unusual Demyelinating Disease of the Central Nervous System Involving Bilateral Corticospinal Tracts Following COVID-19 Infection: A Case Report.](#)

Ju H, Kim YH, Seok JM, Kim BJ.
J Clin Neurol. 2023 Sep;19(5):503-505. doi: 10.3988/jcn.2023.0121.
PMID: 37635429 [Free PMC article.](#) [No abstract available.](#)

[Post-COVID-19 Vaccination CNS Magnetic Resonance Imaging Findings: A Systematic Review.](#)

Ghaderi S, Mohammadi S, Heidari M, Sharif Jalali SS, Mohammadi M.
Can J Infect Dis Med Microbiol. 2023 Jun 29;2023:1570830. doi: 10.1155/2023/1570830. eCollection 2023.
PMID: 37427078 [Free PMC article.](#) [Review.](#)

Publication types

> [Case Reports](#)

MeSH terms

- > [Adult](#)
- > [Brain* / diagnostic imaging](#)
- > [Brain* / pathology](#)
- > [COVID-19 Vaccines* / adverse effects](#)
- > [COVID-19* / complications](#)
- > [COVID-19* / immunology](#)
- > [ChAdOx1 nCoV-19](#)
- > [Demyelinating Diseases* / chemically induced](#)
- > [Humans](#)
- > [Male](#)
- > [Middle Aged](#)
- > [Neuroimaging](#)
- > [Pyramidal Tracts](#)
- > [Spinal Cord / diagnostic imaging](#)
- > [Spinal Cord / pathology](#)
- > [Vaccination / adverse effects](#)

Substances

- > [BBV152 COVID-19 vaccine](#)
- > [ChAdOX1 nCoV-19](#)
- > [COVID-19 Vaccines](#)

Related information

[MedGen](#)

LinkOut – more resources

Full Text Sources

[Ovid Technologies, Inc.](#)

[Wolters Kluwer](#)

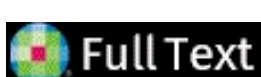
Medical

[MedlinePlus Health Information](#)

Miscellaneous

[NCI CPTAC Assay Portal](#)

FULL TEXT LINKS



ACTIONS

“ Cite

🔖 Collections

SHARE



PAGE NAVIGATION

< [Title & authors](#)

[Abstract](#)

[Conflict of interest statement](#)

[Similar articles](#)

[Cited by](#)

[Publication types](#)

[MeSH terms](#)

[Substances](#)

[Related information](#)

[LinkOut – more resources](#)

< [SARS-CoV-2 vaccine](#) may trigger thyroid **autoimmunity**: real-life experience and review of the literature.

Ruggeri RM, et al. J Endocrinol Invest. 2022. PMID: 35829989

[Free PMC article.](#) [Review.](#)

RESULTS: Our case series gives a description of the full spectrum of **autoimmune** disorders that may occur after SARS-CoV-2 **vaccines** ...

> [New Onset or Deterioration of Thyroid Eye Disease After mRNA SARS-CoV-2 Vaccines](#): Report of 2 Cases and Literature Review.

Mohamed A, et al. J Clin Endocrinol Metab. 2023. PMID: 36251747

[Free PMC article.](#) [Review.](#)

CONTEXT: Occurrence of Graves' disease (GD) has been reported following SARS-CoV-2 **vaccine** administration, but little is known about ...

